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customers = [
    "Beth Johnson",      # customers[0]
    "Teddy Smith",       # customers[1]
    "Greg Holland",      # customers[2]
    "Rebecca Royal",     # customers[3]
    "Andre Jackson",     # customers[4]
    "Vernon Payne",      # customers[5]
    "Kesha Jones",       # customers[6]
    "Maurice Bricks"     # customers[7]
]

customer_orders = [
    (1, customers[4], "Graphic T-Shirt", 14.99, "Clothing"),
    (2, customers[0], "Mouse Pad", 10.49, "Electronics"),
    (3, customers[6], "Ankle Socks", 5.99, "Clothing"),
    (4, customers[2], "Phone Stand", 13.49, "Electronics"),
    (5, customers[1], "Coffee Mug", 8.99, "Home Essentials"),
    (6, customers[5], "Wireless Earbuds", 19.49, "Electronics"),
    (7, customers[3], "Baseball Cap", 17.99, "Clothing"),
    (8, customers[0], "Kitchen Towel Set", 11.99, "Home Essentials"),
    (9, customers[1], "Wireless Earbuds", 19.49, "Electronics"),
    (10, customers[7], "Coffee Mug", 8.99, "Home Essentials"),

    (11, customers[2], "Clothes Hangers", 16.49, "Home Essentials"),
    (12, customers[4], "Coffee Mug", 8.99, "Home Essentials"),
    (13, customers[0], "Baseball Cap", 17.99, "Clothing"),
    (14, customers[5], "Mouse Pad", 10.49, "Electronics"),
    (15, customers[1], "USB-C Cable", 7.49, "Electronics"),
    (16, customers[3], "Mouse Pad", 10.49, "Electronics"),
    (17, customers[6], "Graphic T-Shirt", 14.99, "Clothing"),
    (18, customers[1], "Kitchen Towel Set", 11.99, "Home Essentials"),
    (19, customers[4], "Clothes Hangers", 16.49, "Home Essentials"),
    (20, customers[2], "Coffee Mug", 8.99, "Home Essentials"),

    (21, customers[7], "Kitchen Towel Set", 11.99, "Home Essentials"),
    (22, customers[0], "Graphic T-Shirt", 14.99, "Clothing"),
    (23, customers[3], "Wireless Earbuds", 19.49, "Electronics"),
    (24, customers[1], "Phone Stand", 13.49, "Electronics"),
    (25, customers[5], "USB-C Cable", 7.49, "Electronics"),
    (26, customers[2], "Wireless Earbuds", 19.49, "Electronics"),
    (27, customers[0], "USB-C Cable", 7.49, "Electronics"),
    (28, customers[3], "Graphic T-Shirt", 14.99, "Clothing"),
    (29, customers[1], "Clothes Hangers", 16.49, "Home Essentials"),
    (30, customers[6], "Baseball Cap", 17.99, "Clothing"),

    (31, customers[2], "Kitchen Towel Set", 11.99, "Home Essentials"),
    (32, customers[4], "Ankle Socks", 5.99, "Clothing"),
    (33, customers[0], "Phone Stand", 13.49, "Electronics"),
    (34, customers[1], "Mouse Pad", 10.49, "Electronics"),
    (35, customers[5], "Phone Stand", 13.49, "Electronics"),
    (36, customers[2], "Mouse Pad", 10.49, "Electronics"),
    (37, customers[4], "Baseball Cap", 17.99, "Clothing"),
    (38, customers[0], "Coffee Mug", 8.99, "Home Essentials"),
    (39, customers[3], "Ankle Socks", 5.99, "Clothing"),
    (40, customers[1], "Baseball Cap", 17.99, "Clothing"),

    (41, customers[5], "Coffee Mug", 8.99, "Home Essentials"),
    (42, customers[0], "Wireless Earbuds", 19.49, "Electronics"),
    (43, customers[2], "Baseball Cap", 17.99, "Clothing"),
    (44, customers[0], "Clothes Hangers", 16.49, "Home Essentials"),
    (45, customers[3], "Phone Stand", 13.49, "Electronics"),
    (46, customers[1], "Graphic T-Shirt", 14.99, "Clothing"),
    (47, customers[4], "Kitchen Towel Set", 11.99, "Home Essentials"),
    (48, customers[0], "Ankle Socks", 5.99, "Clothing"),
    (49, customers[2], "Graphic T-Shirt", 14.99, "Clothing"),
    (50, customers[3], "USB-C Cable", 7.49, "Electronics")
]

#Products and Categories
product_categories = {"Home Essentials": ["Coffee Mug", "Kitchen Towel Set", "Clothes Hangers"],
                     "Electronics": ["Mouse Pad", "USB-C Cable", "Phone Stand", "Wireless Earbuds"],
                     "Clothing": ["Baseball Cap", "Ankle Socks", "Graphic T-Shirt"]}
print("Display of all Product categories and Products:".upper())
for name in product_categories.keys():
    print(f"{name}: {' '.join(product_categories[name])}")

#loop to calculate the total amount each customer has spent
print("\nTOTAL REVENUE PER CUSTOMER")

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for customer in customers:
    total_spent = 0
    for order in customer_orders:
        if order[1] == customer:
            total_spent += order[3]
            if total_spent > 100:
                classification = 'High-value'
            elif total_spent > 50 and total_spent <= 100:
                classification = 'Moderate'
            else:
                classification = 'Low-value'
    print(f"{customer} has spent ${total_spent:.2f} and is a {classification} buyer")

#Dictionary of customer totals
customer_total_dict.update({customer: format(total_spent,".2f")})
#print(customer_total_dict)
print("\n")

print("TOTAL REVENUE BY PRODUCT")
#Product Revenue Totals
for category in product_categories.keys():
    category_total = 0
    for order in customer_orders:
        if order[4] == category:
            category_total += order[3]
    print(f"Revenue for product category {category} is ${category_total:.2f}")
    category_total_dict.update({category: format(category_total,".2f")})

#print("\n Total Revenue per product category stored in a dictionary.\n".upper(),category_total_dict)

#Extracting all unique products
print("\nAll unique products".upper())
unique_products = set()
for order in customer_orders:
    unique_products.add(order[2])
for unique in unique_products:
    print("-",unique)

#List of all customers who purchased Electronics
print("\n List of all customers who purchased Electronics".upper())
unique_electronic_customers = set()
for order in customer_orders:
    if order[4] == "Electronics":
        unique_electronic_customers.add(order[1])
for unique in unique_electronic_customers:
    print("-",unique)

#Top 3 customers who spent the most.
print("\nTHE TOP 3 CUSTOMERS WHO SPENT THE MOST ARE:")
top_three_spenders = sorted(customer_total_dict.items(), key=lambda x: float(x[1]), reverse=True)[:3]
for top_three in top_three_spenders:
    print(f"{top_three[0]} spent ${top_three[1]}")

# Customers who purchased from more than one Product Category
print("\nCustomers who purchased from more than 1 product category:".upper())
multiple_category_customers = set()
for customer, categories in customer_category_count.items():
    if len(categories) > 1:
        multiple_category_customers.add(customer)
    print(f"{customer} purchased from {' , '.join(categories)}")

# Customers who bought from both Electronics and Clothing
electronics_customers = set()
clothing_customers = set()
for order in customer_orders:
    if order[4] == "Electronics":
        electronics_customers.add(order[1])
    elif order[4] == "Clothing":
        clothing_customers.add(order[1])
print("\nCustomers who purchased from both Electronics and Clothing:\n".upper(),electronics_customers & clothing_customers)
print()

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DISPLAY OF ALL PRODUCT CATEGORIES AND PRODUCTS:
 Home Essentials: Coffee Mug, Kitchen Towel Set, Clothes Hangers
 Electronics: Mouse Pad, USB-C Cable, Phone Stand, Wireless Earbuds
 Clothing: Baseball Cap, Ankle Socks, Graphic T-Shirt

TOTAL REVENUE PER CUSTOMER

Beth Johnson has spent \$127.40 and is a High-value buyer
Teddy Smith has spent \$121.41 and is a High-value buyer
Greg Holland has spent \$113.92 and is a High-value buyer
Rebecca Royal has spent \$89.93 and is a Moderate buyer
Andre Jackson has spent \$76.44 and is a Moderate buyer
Vernon Payne has spent \$59.95 and is a Moderate buyer
Kesha Jones has spent \$38.97 and is a Low-value buyer
Maurice Bricks has spent \$20.98 and is a Low-value buyer

TOTAL REVENUE BY PRODUCT

Revenue for product category Home Essentials is \$179.85
Revenue for product category Electronics is \$247.31
Revenue for product category Clothing is \$221.84

ALL UNIQUE PRODUCTS

- Kitchen Towel Set
- Baseball Cap
- Clothes Hangers
- Coffee Mug
- USB-C Cable
- Phone Stand
- Mouse Pad
- Wireless Earbuds
- Graphic T-Shirt
- Ankle Socks

LIST OF ALL CUSTOMERS WHO PURCHASED ELECTRONICS

- Greg Holland
- Rebecca Royal
- Vernon Payne
- Teddy Smith
- Beth Johnson

THE TOP 3 CUSTOMERS WHO SPENT THE MOST ARE:

Beth Johnson spent \$127.40
Teddy Smith spent \$121.41
Greg Holland spent \$113.92

CUSTOMERS WHO PURCHASED FROM MORE THAN 1 PRODUCT CATEGORY:

Andre Jackson purchased from Clothing, Home Essentials
Beth Johnson purchased from Electronics, Clothing, Home Essentials
Greg Holland purchased from Electronics, Clothing, Home Essentials
Teddy Smith purchased from Electronics, Clothing, Home Essentials
Vernon Payne purchased from Electronics, Home Essentials
Rebecca Royal purchased from Electronics, Clothing

CUSTOMERS WHO PURCHASED FROM BOTH ELECTRONICS AND CLOTHING:

{'Greg Holland', 'Beth Johnson', 'Rebecca Royal', 'Teddy Smith'}